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HEALTHY LIVING

Osteoporosis

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Osteoporosis is a condition in which a decrease in the density of bones weakens the bones, making breaks (fractures) likely.

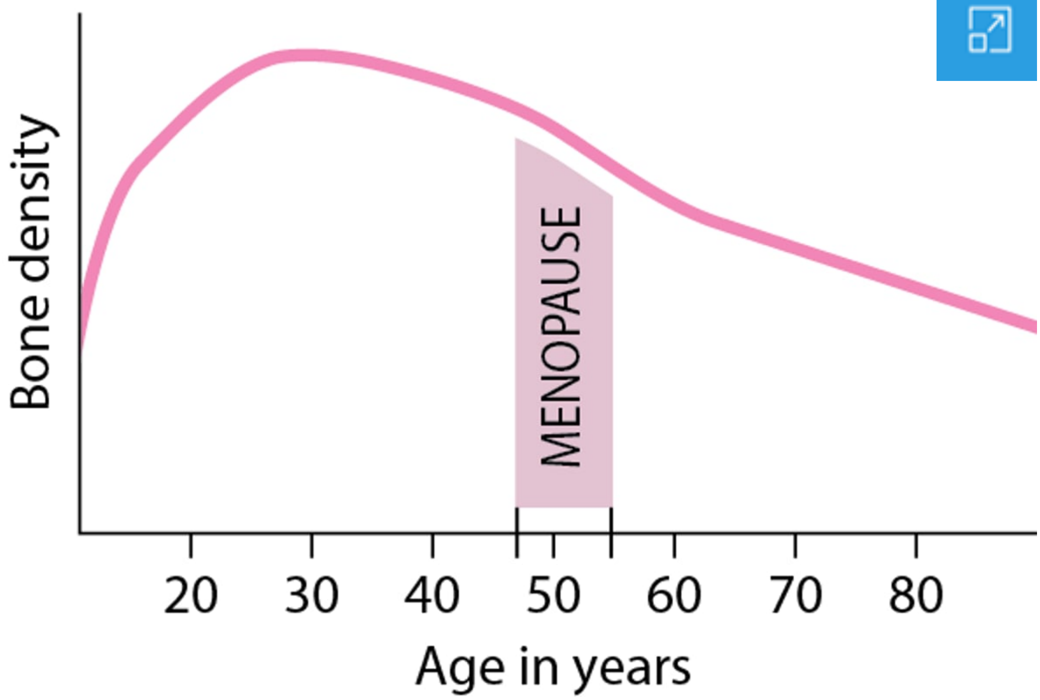
- Aging, estrogen deficiency, low vitamin D or calcium intake, and certain disorders can decrease the amounts of the components that maintain bone density and strength.
- Osteoporosis may not cause symptoms until a bone fracture occurs.
- Fractures can occur with little or no force and may occur after a minor fall.
- Although fractures are often painful, some fractures of the spine do not cause pain but can cause deformities.
- Doctors diagnose people at risk by testing their bone density.
- Osteoporosis can usually be prevented and treated by managing risk factors, ensuring adequate calcium and vitamin D intake, engaging in weight-bearing exercise, and taking bisphosphonates or other medications.

Bones contain minerals, including calcium and phosphorus, which make them hard and dense. To maintain bone density (or bone mass), the body requires an adequate supply of calcium and other minerals and must produce the proper amounts of several hormones, such as parathyroid hormone, growth hormone, calcitonin, estrogen, and testosterone. An adequate supply of vitamin D is needed to absorb calcium from food and incorporate it into bones. Vitamin D is absorbed from the diet and also manufactured in the skin using sunlight.

So that bones can adjust to the changing demands placed on them, they are continuously broken down and reformed. This process is known as remodeling. In this process, small areas of bone tissue are continuously removed and new bone tissue is deposited. Remodeling affects the shape and density of the bones.

Loss of Bone Density in Women

In women, bone density (or mass) progressively increases until about age 30, at which time bones are at their strongest. After that, bone density gradually decreases. The decrease in bone density accelerates after menopause, which occurs on average around age 51.



Because more bone is formed than is broken down in the young adult years, bones progressively increase in density until about age 30, when they are at their strongest. After that, as breakdown exceeds formation, bones slowly decrease in density. If the body is unable to maintain an adequate amount of bone formation, bones continue to lose density and may become increasingly fragile, eventually resulting in osteoporosis.

Types of Osteoporosis

Osteoporosis is more common in women. It affects almost 20% (1 in 5) of women aged 50 and over and almost 5% (1 in 20) of men aged 50 and over. Approximately

50% of postmenopausal women and 20% of men over age 50 years will sustain an osteoporosis-related fracture in their lifetimes. There are two main types of osteoporosis:

- Primary osteoporosis: Occurs spontaneously
- Secondary osteoporosis: Is caused by another disorder or by a medication

Primary osteoporosis

Nearly all occurrences of osteoporosis in both men and women are primary. Most cases occur in postmenopausal women and in older men.

A major cause of osteoporosis is a lack of estrogen, particularly the rapid decrease that occurs at menopause. Most men over 50 have higher estrogen levels than postmenopausal women, but these levels also decline with aging, and low estrogen levels are associated with osteoporosis in both men and women. Estrogen deficiency increases bone breakdown and results in rapid bone loss. In men, low levels of male sex hormones also contribute to osteoporosis. Bone loss is even greater if calcium intake or vitamin D levels are low. Low vitamin D levels result in calcium deficiency, and increased activity of the parathyroid glands causes the glands to release too much parathyroid hormone (see hyperparathyroidism), which can also stimulate bone breakdown. Bone production also decreases.

A number of other factors, such as certain medications, tobacco use, heavy alcohol use, a family history of osteoporosis (for example, if a person's mother or father has had a hip fracture), and small body stature, increase the risk of bone loss and the development of osteoporosis in women. These risk factors are also important in men.

Secondary osteoporosis

Examples of disorders that may cause secondary osteoporosis are chronic kidney disease, and hormonal disorders (especially Cushing disease, hyperparathyroidism, hyperthyroidism, hypogonadism, high levels of prolactin, and diabetes mellitus). Certain types of cancer, such as multiple myeloma, can cause secondary osteoporosis, as can other diseases such as celiac disease and rheumatoid arthritis. Examples of medications that, if used for a long time, may cause secondary osteoporosis are progesterone, steroids (also sometimes referred to as glucocorticoids or corticosteroids), thyroid hormones, aromatase inhibitors (for breast cancer), and antiseizure medications. Excessive alcohol and cigarette smoking may contribute to osteoporosis.

Risk Factors for Primary Osteoporosis

- Family members with osteoporosis
- A diet that is low in calcium and vitamin D
- Sedentary lifestyle
- Thin build
- Early menopause
- Cigarette smoking
- Excessive alcohol consumption

Idiopathic osteoporosis

Idiopathic osteoporosis is a rare type of osteoporosis. The word *idiopathic* simply means that the cause is unknown. This type of osteoporosis occurs in premenopausal women, in men under age 50, and in children and adolescents who have normal hormone levels, normal vitamin D levels, and no obvious reason to have weak bones.

Symptoms of Osteoporosis

At first, osteoporosis causes no symptoms. Bone density loss occurs very gradually. Some people never develop symptoms. However, when osteoporosis causes bones to break (fracture), people may have pain depending on the location of the fracture. Fractures tend to heal slowly in people who have osteoporosis and may lead to deformities such as curvature of the spine.

In long bones, such as the bones of the arms and legs, the fracture usually occurs at the ends of the bones rather than in the middle. Long bone fractures typically are painful.

The bones of the spine (vertebrae) are particularly at risk of fracture due to osteoporosis. These fractures are the most common osteoporosis-related fracture. They usually occur in the middle to lower back. Typically, the drum-shaped body of one or more vertebrae collapses into itself and becomes compressed into a wedge shape. These vertebral compression fractures may occur in people who have any type of osteoporosis, including those on medications that cause bone density loss and increased risk of fracture. The weakened vertebrae may collapse spontaneously or after a slight injury, such as with coughing or with a fall from a standing height.

Most of these vertebral compression fractures do not cause pain. However, pain can develop, usually starting suddenly, staying in a particular area of the back, and worsening when a person stands or walks. The area may be tender. Usually the pain and tenderness begin to go away gradually after 1 week. However, lingering pain may last for months or be constant. If several vertebrae break, an abnormal curvature of the spine may develop, causing muscle strain and soreness as well as deformity.

Compression Fracture Due to Osteoporosis

The dark line just below the arrow represents a fracture of a lumbar vertebra (in the lower back). This vertebra is compressed, and so has less height than it normally would have.

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Fragility fractures are fractures that result from a relatively minor strain or fall, such as a fall from a standing height or less, including a fall out of bed, that normally would not cause a fracture in a healthy bone. Fragility fractures commonly occur in the wrist, hip, and spine (vertebral compression fractures). Other bones include the upper arm bone (humerus) and the pelvis.

Hip fracture, one of the most serious fractures, is a major cause of disability and loss of independence in older adults, and is associated with reduced survival.

Wrist fractures occur often, especially in people with postmenopausal osteoporosis.

People who have had one fracture in which osteoporosis had been a factor are at much higher risk of having more such fractures.

Fractures of the nose, ribs, collarbone, knee cap, and bones in the feet are not considered osteoporosis-related fractures.

Did You Know...

- People who have had one osteoporosis-related fracture are at much higher risk of having more of these fractures.

Diagnosis of Osteoporosis

- Bone density testing
- A fracture caused by something that would not normally cause a fracture, such as falling from a standing height or out of bed (called a fragility fracture)

A doctor may suspect osteoporosis in the following people:

- All women age 65 or older
- Women between menopause and age 65 who have risk factors for osteoporosis
- All men and women who have had a previous fracture caused by little or no force, even if the fracture occurred at a young age
- People whose bones appear less dense on x-rays or who have vertebral compression fractures on x-rays
- People who are at risk of developing secondary osteoporosis

If osteoporosis is suspected and people have not had x-rays, doctors may order imaging to diagnose a fracture. Certain findings on x-rays suggest osteoporosis, but the diagnosis of osteoporosis is confirmed by bone density testing.

Bone density testing

Bone density testing can be used to detect or confirm suspected osteoporosis, even before a fracture occurs.

Dual-energy x-ray absorptiometry (DXA scan) is the most useful test of bone density. The DXA scan takes high-energy and low-energy x-rays of the spine and hip, which are the sites at which major fractures are likely to occur. The difference between the high- and low-energy x-ray readings allows doctors to calculate bone density. The result is reported as a T-score, which compares a person's bone density to the density of a healthy person of the same sex and race/ethnicity at the age of peak bone mass, which is about age 30. The lower the bone density, the lower the T-score. A T-score of -2.5 or lower defines osteoporosis.

DXA scans are painless, involve very little radiation, and can be completed in about 10 to 15 minutes. They may be useful for monitoring the response to treatment as well as for making the diagnosis of osteoporosis. DXA scans may also reveal osteopenia, a condition in which bone density is decreased but not as severely as in osteoporosis. People who have osteopenia also have an increased risk of fractures. It is important for your doctor to calculate a fracture risk assessment (FRAX) score if you have osteopenia on DXA scan, to provide an estimate of your risk for fracture.

People who are already taking osteoporosis medications (eg, a bisphosphonate, an anabolic agent, denosumab) should have repeat DXA scans to monitor the effectiveness of the treatment.

Osteoporosis can also be diagnosed in a person with a fracture from something that would not normally cause a fracture (known as a fragility fracture), even if their bone density test is normal.

Other tests

Blood tests may be done to measure calcium, vitamin D, and levels of certain hormones.

Further testing may be needed to rule out treatable conditions that might lead to osteoporosis. If such a condition is found, the diagnosis is called secondary osteoporosis.

Treatment of Osteoporosis

- Calcium and vitamin D
- Risk factor modification
- Weight-bearing exercise
- Medications
- Treatment of fractures

Osteoporosis treatment involves risk factor modification, which includes ensuring adequate intake of calcium and vitamin D and engaging in weight-bearing exercises (such as walking, climbing stairs, or weight training, all of which strengthen bones). Other lifestyle modifications, such as avoidance of tobacco or excess alcohol intake as well as fall prevention, are also important strategies to reduce the risk of fracture. Treatment with medications is usually recommended. When treating people who have osteoporosis, doctors also manage conditions and risk factors that can contribute to ongoing bone loss.

Calcium and vitamin D

Consuming an adequate amount of nutrients, particularly calcium and vitamin D, is helpful, especially before maximum bone density is reached (around age 30) but also after this time. Vitamin D helps the body absorb calcium.

All men and women should consume at least 1,000 milligrams of calcium each day. Postmenopausal women, older men, children who are going through puberty, and women who are pregnant or breastfeeding may need to consume 1,200 to 1,500 milligrams each day. Calcium in food is preferred over calcium supplements. Foods rich in calcium include dairy products (such as milk and yogurt), certain vegetables (such as broccoli), nut milks (such as almond milk), and nuts (such as macadamia). See table Amount of Calcium in Some Foods.

However, if people cannot consume the recommended amounts by diet alone, they need to take supplements. Many calcium preparations are available, and some include supplemental vitamin D. The most common supplements are calcium carbonate or calcium citrate. Calcium citrate supplements should be taken by people who take a gastric acid suppressant (for example, an H2 blocker, such as famotidine, or proton pump inhibitor, such as omeprazole, which are used to reduce stomach acid production) or who have had gastric bypass surgery.

Amount of Calcium in Some Foods

Food	Serving Size	Approximate Amount of Calcium (in milligrams)
Yogurt, plain, low fat	8 ounces	450
Yogurt, Greek, plain, low fat	8 ounces	260
Milk, low fat (1 %)	1 cup	310
Milk, fat free (skim)	1 cup	300
Soy beverage (soy milk), unsweetened	1 cup	300
Almond beverage (almond milk), unsweetened	1 cup	440
Rice beverage (rice milk), unsweetened	1 cup	280
Cheese, reduced, low, or fat free (various)	1½ ounce	120-490
Collard greens, cooked	1 cup	270
Spinach, cooked	1 cup	250
Bok choy, cooked	1 cup	190
Kale, cooked	1 cup	180
Chard, cooked	1 cup	100
Tofu, raw, regular, prepared with calcium sulfate	½ cup	430
Sardines, canned	3 ounces	330
Grapefruit juice, 100%, fortified	1 cup	350
Orange juice, 100%, fortified	1 cup	350